MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 6

1. *Determine a basis for each of the following subspaces:*

(i) $H = \left\{ \begin{bmatrix} 4s \\ -3s \\ -t \end{bmatrix} : s, t \in \mathbb{R} \right\}$.

(ii) $K = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \ : \ a - 3b + c = 0 \right\}$.

2. *Find a basis for the following subspaces:*

(i)
$$W_3 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times 2} : a = d = 0 \right\}.$$

(ii)
$$W_4 = \left\{ p(t) \in \mathbb{P}_2 : \frac{d}{dt} p(t) = 0 \right\}.$$

(Recall that $\mathbb{P}_2$ is the vector space of polynomials of degree at most 2).

3. *Let $M_{2\times 2}$ be the vector space of all 2×2 matrices. Consider the two bases*

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

and

$$\mathcal{C} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$$

and $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \in M_{2\times 2}$. *What are $\mathbf{A}_{\mathcal{B}}$ and $\mathbf{A}_{\mathcal{C}}$?*

4. *Let A, B be two 4×3 matrices. Let $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ be the columns of A and let $\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3$ be the columns of B.*

(i) Suppose that $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are linearly independent. Find a basis for $\mathrm{Col}(A)$ and a basis for $\mathrm{Nul}(A)$. What are the dimensions of $\mathrm{Col}(A)$ and $\mathrm{Nul}(A)$?

(ii) Suppose that $\mathbf{b}_1, \mathbf{b}_2$ are linearly independent and $\mathbf{b}_3 = 2\mathbf{b}_1 + 7\mathbf{b}_2$. Find a basis for $\mathrm{Col}(B)$ and a basis for $\mathrm{Nul}(B)$.

5. *True or False? Give arguments why! In the questions below, V is an arbitrary vector space and A is an arbitrary $m \times n$ matrix.*

(1) The number of pivot columns of a matrix equals the dimension of its column space.

(2) A plane in $\mathbb{R}^3$ is a two-dimensional subspace of $\mathbb{R}^3$.

(3) The dimension of the vector space $\mathbb{P}_4$ (the vector space of polynomials of degree at most 4) is 4.

(4) If $\dim V = n$ and S is a linearly independent set in V, then S is a basis for V.

(5) If a set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ spans a vector space V and if T is a set of more than p vectors in V, then T is linearly dependent.

(6) If $\dim V = n$ and if S spans V, then S is a basis of V.

(7) If B is an Echelon Form of A, and if B has three nonzero rows, then the first three rows of A form a basis for the row space $\mathrm{Col}(A^T)$ of A.

(8) The dimensions of the row space and the column space of A are always the same, even if A is not a square matrix.

(9) The sum of the dimensions of the row space and the null space of A equals the number of rows in A.

(10) If B is an echelon form of A, then the pivot columns of B form a basis for the column space of A.

(11) The dimension of the null space of A is the number of columns of A that are *not* pivot columns.

(12) If two matrices A and B are row equivalent, then the vector spaces $\mathrm{Col}(A)$ and $\mathrm{Col}(B)$ have the same dimension.

6. (1) *Construct a* 2×2*-matrix whose nullspace equals its column space.*

(2) Why does no 3×3-matrix have a nullspace that equals its column space?

7. *Construct a matrix with the required property, or explain why such a matrix cannot exist.*

(1) Column space contains $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$, row space contains $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\5\end{bmatrix}$.

(2) Column space has basis $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}\right\}$, nullspace has basis $\left\{\begin{bmatrix}3\\2\\1\end{bmatrix}\right\}$.

(3) Dimension of nullspace = 1 + dimension of left nullspace.

(4) Left nullspace contains $\begin{bmatrix}1\\3\end{bmatrix}$, row space contains $\begin{bmatrix}3\\1\end{bmatrix}$.

8. *Let* $\mathcal{A} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ *be a basis of* $\mathbb{R}^3$.

(1) Find a 3×3-matrix B such that $\mathbf{v} = B\mathbf{v}_{\mathcal{A}}$ for all $\mathbf{v} \in \mathbb{R}^3$.

(2) Find a 3×3-matrix C such that $\mathbf{v}_{\mathcal{A}} = C\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^3$.

9. *Consider the following eight vectors in* $\mathbb{R}^8$*:*

$$\mathcal{B} = (\begin{bmatrix}1\\1\\0\\0\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\1\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\0\\0\\1\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\0\\0\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\-1\\1\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\-1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\0\\0\\-1\\1\end{bmatrix}), \ \boldsymbol{w} = \begin{bmatrix}100\\200\\44\\50\\20\\20\\4\\2\end{bmatrix}.$$

(1) Show that $\mathcal{B}$ is a basis of $\mathbb{R}^8$.

(2) Calculate $\boldsymbol{w}_{\mathcal{B}}$.

(3) Suppose Alice needs to inform Bob about $\boldsymbol{w}$. However, Alice can transmit only four non-zero numbers (and not eight) to Bob; that is Alice can only send Bob a vector in $\mathbb{R}^8$ which has at most four non-zero entries. Luckily Bob does not need to know the exact values in $\boldsymbol{w}$, a good approximation suffices. Using your result in (2), which vector should Alice communicate to Bob?

10. *Let* $T : \mathbb{R}^2 \to \mathbb{R}^3$ *be the linear transformation with*

$$T\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) = \begin{bmatrix}5\\0\\1\end{bmatrix}, \quad T\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = \begin{bmatrix}0\\1\\0\end{bmatrix}.$$

(1) Consider the basis $\mathcal{B}_1 = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$ of $\mathbb{R}^2$ and the basis $\mathcal{B}_2 = \left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ of $\mathbb{R}^3$. Determine $T_{\mathcal{B}_2,\mathcal{B}_1}$.

(2) Is $T(\mathbf{v}) = T_{\mathcal{B}_2\mathcal{B}_1}\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^2$?

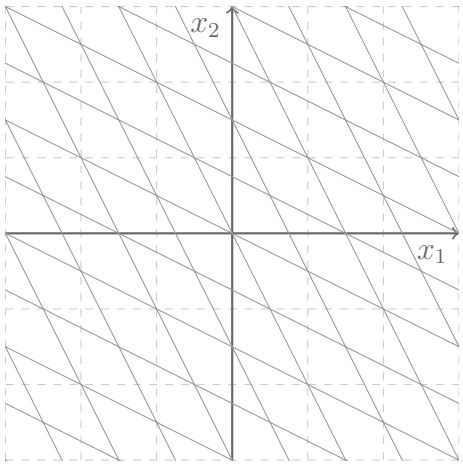
(3) Consider the basis $\mathcal{C}_1 := \left\{\begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}$ of $\mathbb{R}^2$ and the basis $\mathcal{C}_2 = \left\{\begin{bmatrix}5\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ of $\mathbb{R}^3$. Determine $T_{\mathcal{C}_2,\mathcal{C}_1}$.

(4) Is $T(\mathbf{v}) = T_{\mathcal{C}_2,\mathcal{C}_1}\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^2$?

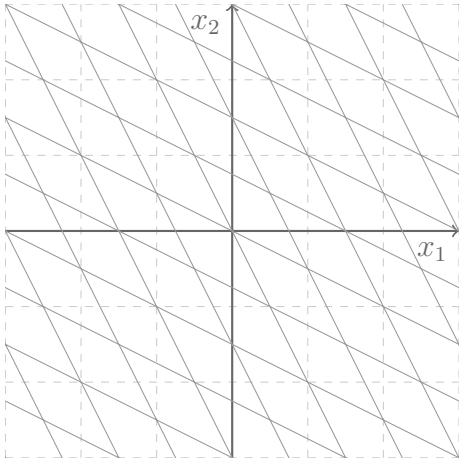
(5) Is $T(\mathbf{v})_{\mathcal{C}_2} = T_{\mathcal{C}_2,\mathcal{C}_1}\mathbf{v}_{\mathcal{C}_1}$ for all $\mathbf{v} \in \mathbb{R}^2$?

11. *Consider the following basis* $\mathcal{B} := \left(\mathbf{b}_1 = \begin{bmatrix}1\\-0.5\end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix}-0.5\\1\end{bmatrix}\right)$ *of* $\mathbb{R}^2$*. Let* $\mathbf{v} = 3\mathbf{b}_1 + 2\mathbf{b}_2$.

(i) *Draw* $\mathbf{b_1}, \mathbf{b_2}$ *and* $\mathbf{v}$ *in the graph below.*



(ii) *Consider a linear transformation* $T : \mathbb{R}^2 \to \mathbb{R}^2$ *such that* $T_{\mathcal{BB}} = \begin{bmatrix} -\frac{2}{3} & 0 \\ 0 & 0.5 \end{bmatrix}$. *Draw* $T(\mathbf{v})$.



12. *For each of the following functions, determine if they are linear transformations.*

(1) $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$

(2) $\det : M_{2\times 2} \to \mathbb{R}$ such that $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$, where $M_{2\times 2}$ is the vector space of all 2×2 matrices.

(3) $\text{tr} : M_{2\times 2} \to \mathbb{R}$ such that $\text{tr}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a + d$, where $M_{2\times 2}$ is the vector space of all 2×2 matrices.

(4) $T : M_{3\times 3} \to M_{3\times 3}$ such that $T\left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & a & b \\ 0 & d & e \end{bmatrix}$ where $M_{3\times 3}$ is the vector space of all 3×3 matrices.

13. *True or False? Justify your answers.*

(1) Let $T : V \to W$ be a linear transformation and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be vectors in V. If $T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)$ are linearly independent, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are also linearly independent.

(2) Let $T : V \to W$ be a linear transformation and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be vectors in V. If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent, then $T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)$ are also linearly independent.

(3) Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span V and let $T : V \to W$. Do the images $T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)$ span W?

14. *This problem explores the connection between linear algebra and quantum mechanics. In particular, we interpret in a particular case, the so called* ***Heisenberg's uncertainty principle*** *which can be stated as follows: It is impossible to have absolute precision while measuring simultaneously the position and momentum of a quantum particle. In terms of linear transformations, this principle asserts that the linear transformations associated to position and momentum do not commute.*

(1) **(The position operator).**Consider the case in which the spaces of quantum states are vector spaces of polynomials P_k (of degree at most k). Consider the linear transformation $\hat{x} : P_{n-1} \to P_n$ defined as

$$\begin{aligned} \hat{x} : P_{n-1} &\to P_n \\ p(t) &\mapsto tp(t). \end{aligned}$$

For instance $\hat{x}(t^2 - t + 2) = t^3 - t^2 + 2t$. Check that this is indeed a linear transformation and find its matrix representation $\hat{X}$ with respect to the standard basis of P_{n-1} and P_n.

(2) **(The momentum operator).** Now consider the transformation $\hat{q} : P_n \to P_{n-1}$ defined as

$$\begin{aligned} \hat{q} : P_n &\to P_{n-1} \\ p(t) &\mapsto \frac{d}{dt}p(t). \end{aligned}$$

For instance $\hat{q}(t^3 - t^2 + 2) = 3t^2 - 2t$. Check that this is indeed a linear transformation and find its matrix representation $\hat{Q}$ with respect to the standard basis of P_n and P_{n-1}.

(3) **(The uncertainty principle).**Let $p(t)$ be a polynomial. Check that

$$\hat{x}\hat{q}(p(t)) \neq \hat{q}\hat{x}(p(t)).$$

This means that the position and momentum operators do not commute!

(4) **(The Heisenberg relation).** Now consider that the space of quantum states is the space $F(\mathbb{R})$ of real functions. The position and momentum operators are defined as above. The commutator $[\hat{q}, \hat{x}]$ of $\hat{q}$ and $\hat{x}$ is defined to be the linear transformation $\hat{q}\hat{x} - \hat{x}\hat{q} : F(\mathbb{R}) \to F(\mathbb{R})$. Use your knowledge of calculus to check that, for every function f:

$$[\hat{q}, \hat{x}]f = f.$$

This is another instance of the Heisenberg uncertainty principle!